ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems
MATLAB Examples

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Term 3, 2026

Topic 2: Discrete-Time Systems – impulse response

Example 1

Given the following difference equation $y[n] = x[n] + y[n-1] - 0.9y[n-2]$,

a. Calculate and plot the impulse response $h[n]$ at $n = 0, \ldots, 120$

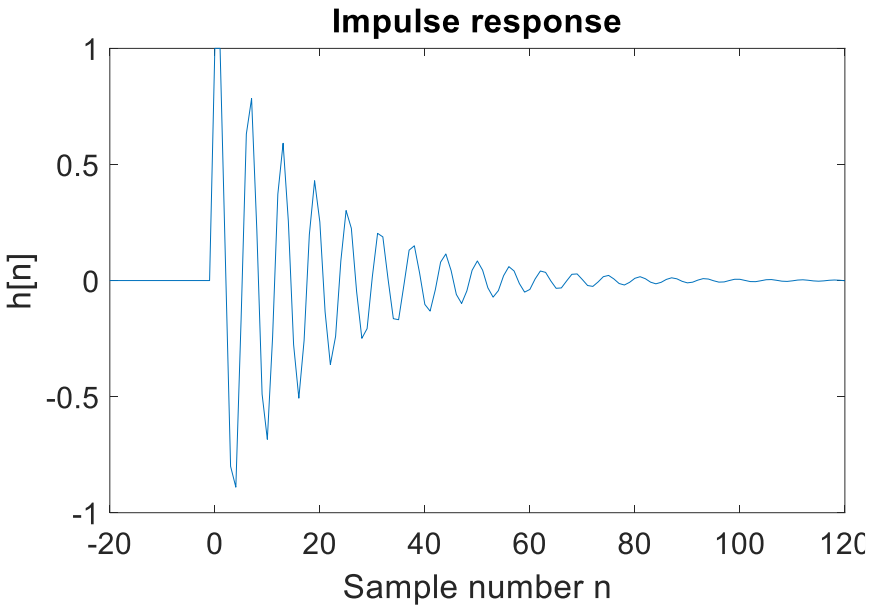
b. Is the system specified by $h[n]$ stable?

Note: Rearrange the difference equation as: $y[n] - y[n-1] + 0.9y[n-2] = x[n]$

From the rearranged difference equation, the numerator coefficients array **b** = [1] and the denominator coefficients array **a** = [1, -1, 0.9]

To compute the impulse response, MATLAB provides the function impz **(b,a,**n**)** and n is the sample number

```
% Impulse response calculation
b = 1; % numerator coefficients
a = [1, -1, 0.9]; % denominator coefficients
n = 0:120; % sample numbers
h = impz(b,a,n); % calculating impulse response of a system
figure
plot(n, h);
title('Impulse response');
xlabel('Sample number n'); ylabel('h[n]');
sum_imp = sum(abs(h)); % sum = 14.8785; Hence, stable system
```



Stability (Page 8): To determine the stability of a system , we need to sum the |h[n]| and if the sum converges to a number, which implies a stable system. We can find $\sum |h[n]|$ using the MATLAB command 'sum' = 14.8785

We also can see that $h[n]$ is practically zero for n >120, therefore the system is stable.

Topic 2: Discrete-Time Systems – Convolution

Example 2

Given the following two finite duration sequences:

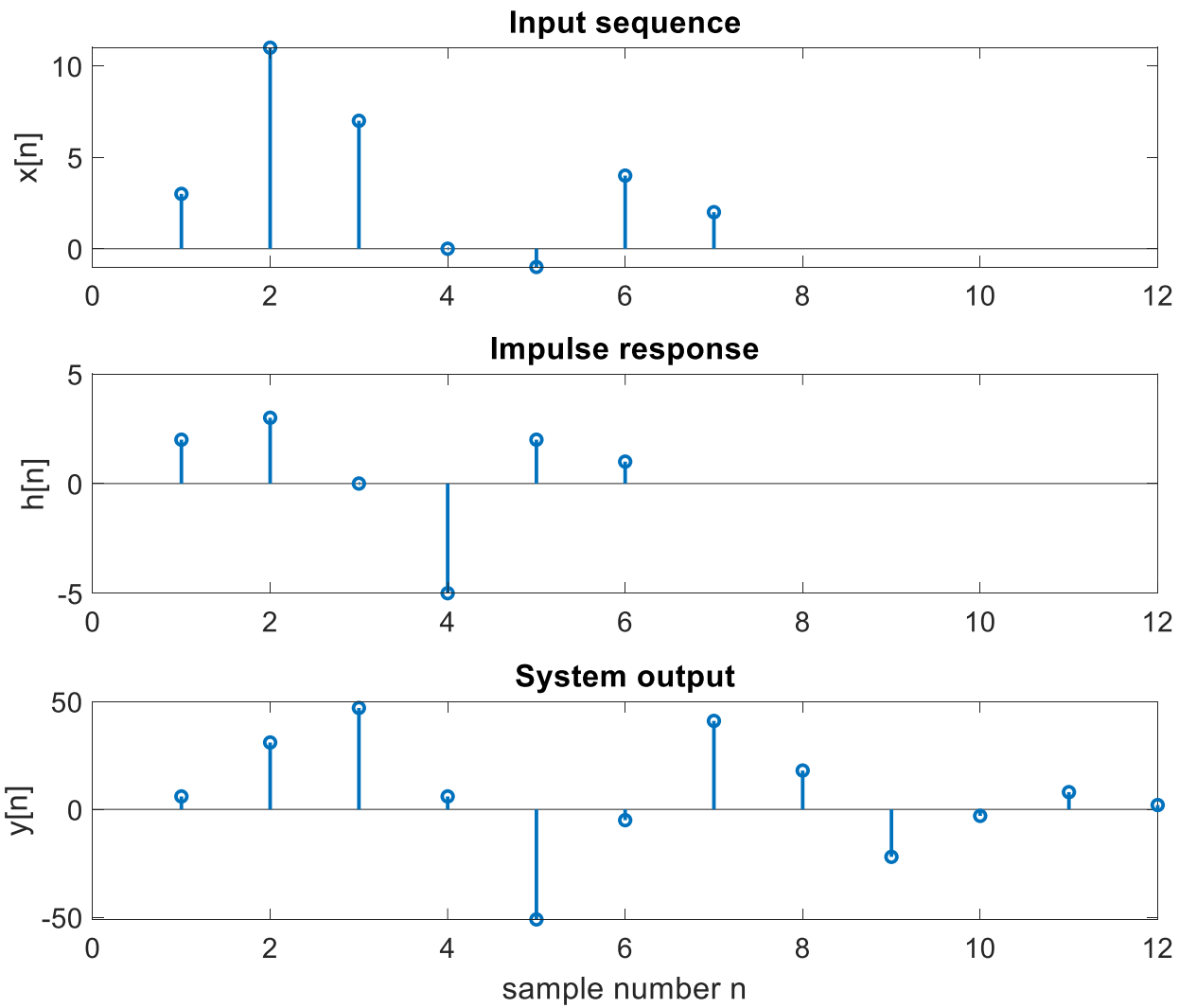
Input sequence length N: $x[n] = [3, 11, 7, 0, -1, 4, 2];$

Impulse response length L: $h[n] = [2, 3, 0, -5, 2, 1];$

determine the convolution y[n] = x[n] * h[n]. $x[n] \rightarrow h[n] \rightarrow y[n]$

Note: MATLAB provides a function called 'conv' that computes the convolution between two finite duration sequences. The 'conv' assumes that the two sequences begin at $n = 0$ as shown by the arrows above. The resultant sequence length is N + L - 1.

```
% convolution calculation
x = [3, 11, 7, 0, -1, 4, 2]; % input sequence length N=7
h = [2, 3, 0, -5, 2, 1]; % Impulse response L=6
y = conv(x, h); % compute the convolution
% y = 6, 31, 47, 6, -51, -5, 41, 18, -22, -3, 8, 2
len = length(y); % length of the sequence y = N+L-1 = 12
figure
subplot 311; stem(x); xlim([0,12]); ylabel('x[n]');
subplot 312; stem(h); xlim([0,12]); ylabel('h[n]');
subplot 313; stem(y); xlim([0, 12]);
xlabel('Sample number n'); ylabel('y[n]');
```



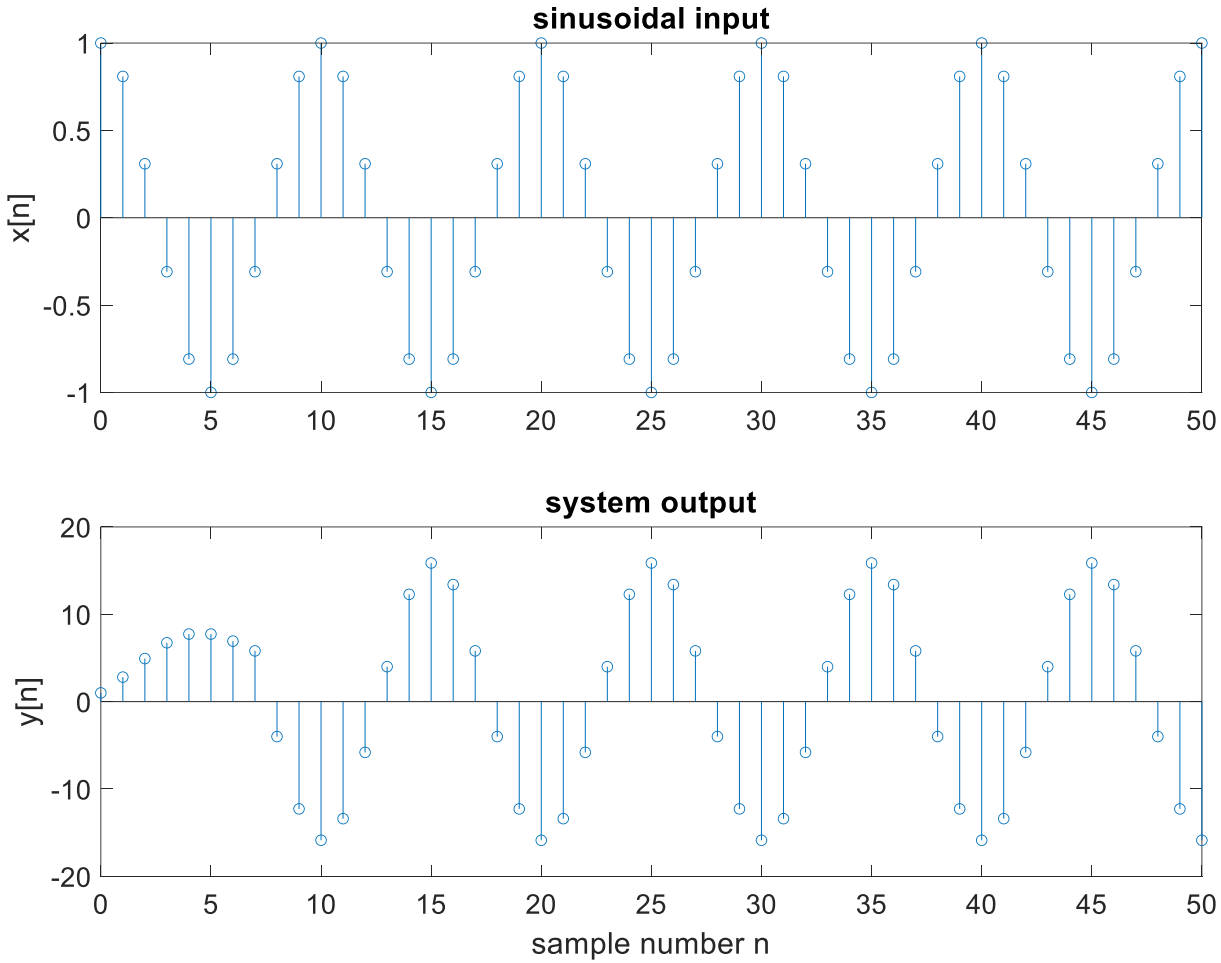
Topic 2: Discrete-Time Systems – Convolution

Example 3

A sinusoid input $x[n] = \cos(0.2\pi n)\ \ 0 \le n \le 50$ is given to a LTI system whose impulse response is described by $h[n] = [\underset{\downarrow}{1}\ 2, 3, 4, 5, 6, 7, 8]$; Find the output y[n] of the system and plot the result.

Note: The period of x[n] is N=10. You will notice that the output of the LTI system will be a sinusoid at the input frequency.

```
% convolution calculation
n = 0:50; % create the sample index
x = cos(0.2*pi*n); % signal x
h = 1:8; % signal h
y = conv(x,h);
y = y(1:length(n)); % truncate to length of x
figure
subplot 211; stem(n, x); % plot input
ylabel('x[n]'); title('sinusoidal input');
subplot 212; stem(n, y); % plot output
xlabel('Sample number n'); ylabel('y[n]');
title('system output')
```



Comments: From the plots, you would observe the periodic nature of the response except for start-up transients. You would also be able to see the period of both the input and output as N = 10.The period of x[n] is N=10.

Topic 2: Discrete-Time Systems – Pole-Zero Description (page 26)

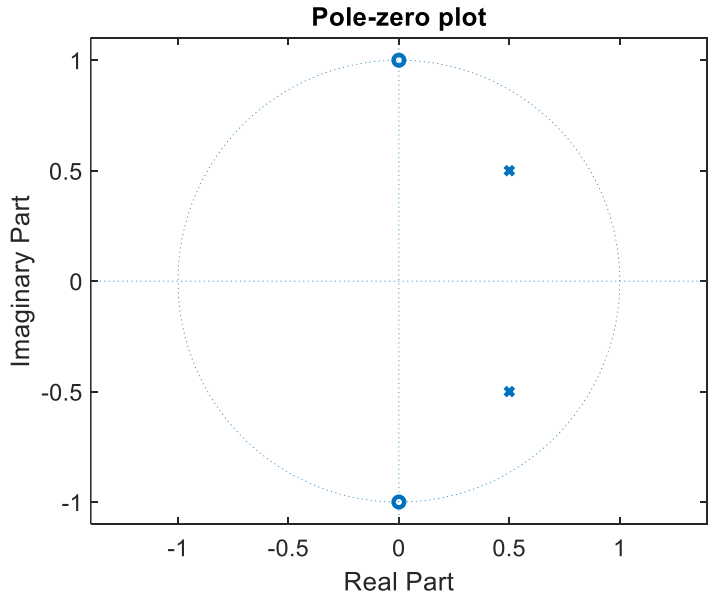
Example 4

Given a causal system $H(z) = \frac{(1+z^{-2})}{(1-z^{-1}+0.5z^{-2})}$

Sketch its pole-zero plot using MATLAB.

Note: By using the 'zplane' function in MATLAB, we can display poles and zeros of a discrete-time system.

```
% pole-zero plot
b = [1, 0, 1]; % numerator coefficient
a = [1, -1, 0.5]; % denominator coefficient
figure; zplane (b, a)% pole-zero plot
title('Pole-zero plot')
```



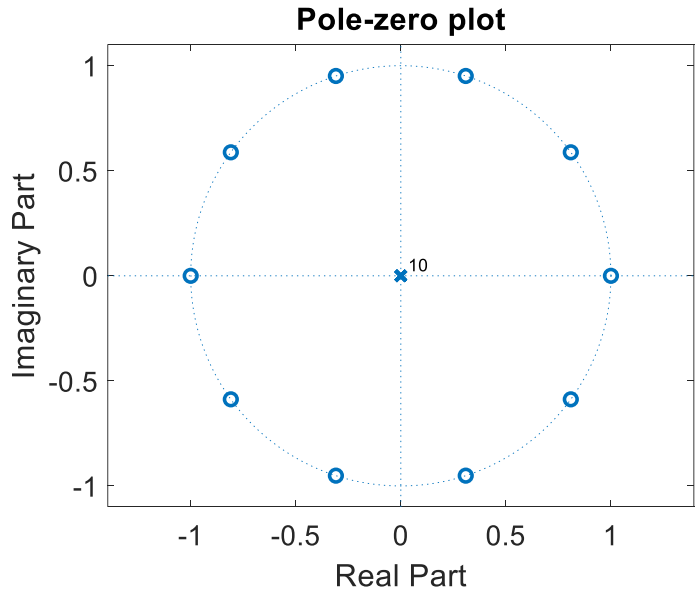
Example 5

A 10th order comb filter has the system function $H(z) = 1 - z^{-10}$.

Determine the pole-zero locations of H(z).

$H(z) = 1 - z^{-10}$
$= (z^{10} - 1)/z^{10}$

```
% pole-zero plot
b = [1, 0, 0, 0, 0, 0, 0, 0, 0, 0,-1]; % numerator coefficient
a = 1; % denominator coefficient
figure; zplane (b, a)% pole-zero plot
title('Pole-zero plot')
```

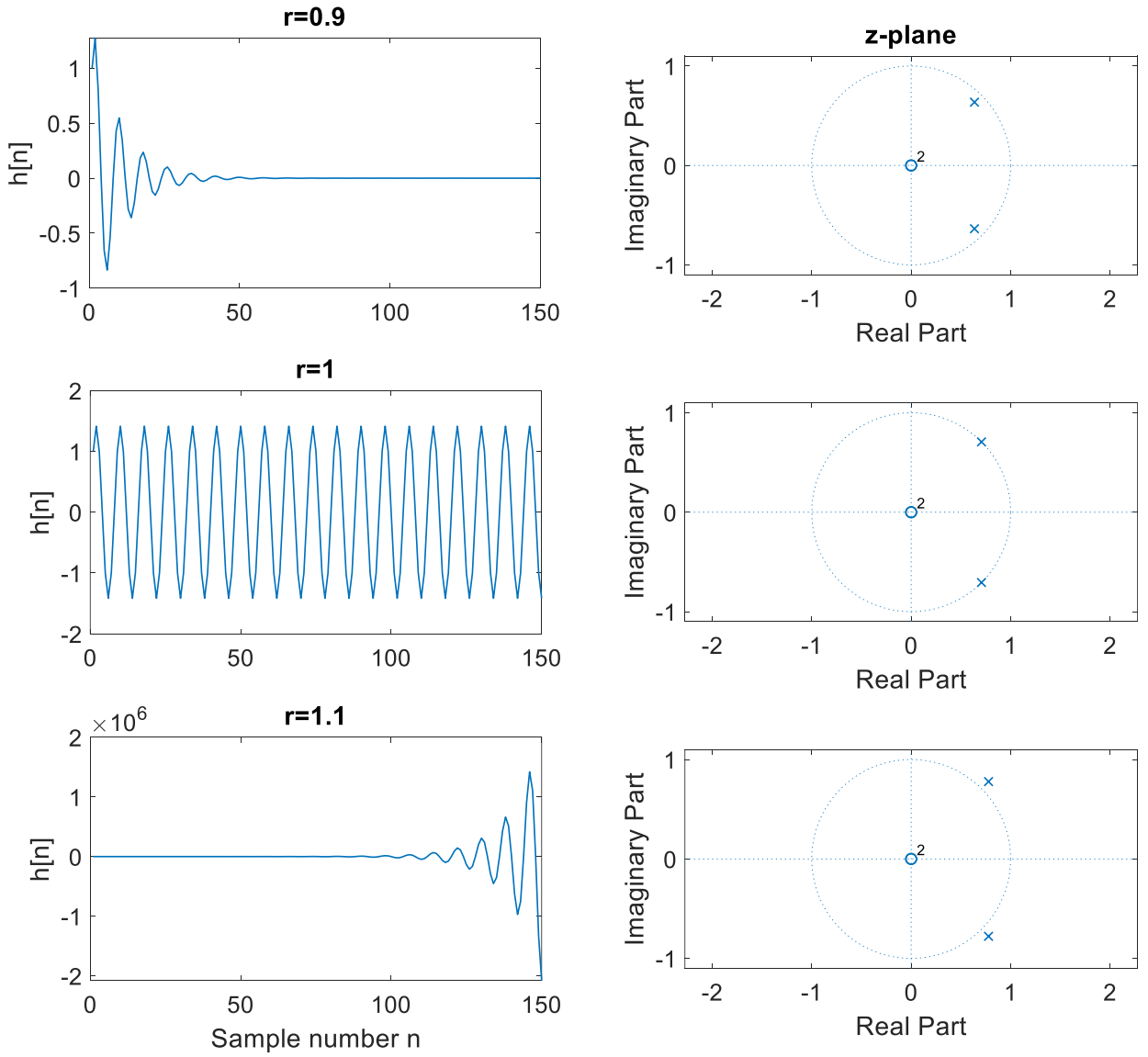


Topic 2: Discrete-Time Systems – Pole-Zero Description

Example 6

A 2nd order system has two zeros at the origin and two poles as $p1 = re^{j\frac{\pi}{4}}$, $p2 = re^{j\frac{-\pi}{4}}$, described by the following transfer function: $H(z) = \frac{z^2}{(z-p_1)(z-p_2)}$. Write a MATLAB program to compute and plot the impulse response, and pole-zero plane (z-plane) of $H(z)$ for the following r values: (a) r = 0.9, (b) r = 1 and (c) r = 1.1.

```
theta = pi/4; % pole position
b = 1; % numerator coefficient
r = [0.9, 1, 1.1]; % radius of the circle
N = 150; % number of samples
figure
% for loop: loop through all r values
for i=1:length(r)
        % denominator coefficient
        a = [1 -2*r(i)*cos(theta) r(i)^2];
        subplot(3, 2, 1+(i-1)*2)
        plot(impz(b,a,N)); % impulse response
        ylabel('h[n]');
        title(['r = ' num2str(r(i))]);
        subplot(3, 2, 2+(i-1)*2); zplane(b,a); % pole-zero plot
end
```



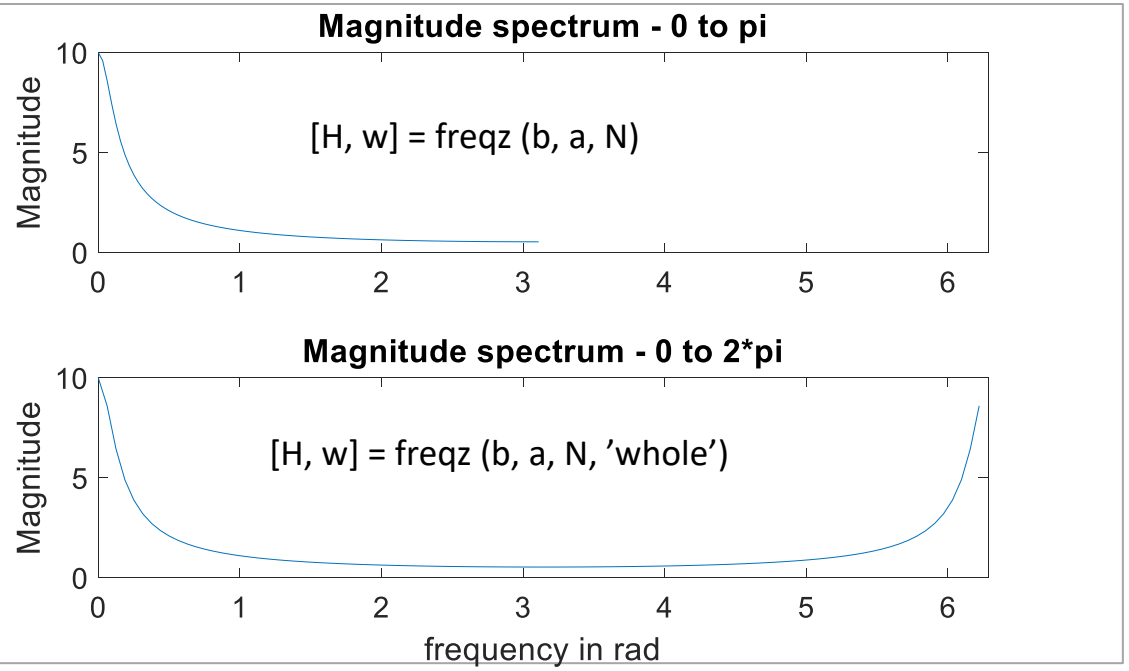
Topic 2: DTFT - Discrete Time Fourier Transform

Example 7

Given a causal system $H(z) = \frac{1}{1-0.9z^{-1}}$. Determine $|H(\theta)|$ using MATLAB function “freqz”.

Note: MATLAB provides the function “freqz” for calculating frequency response (H) of a transfer function. The function is invoked by [H, w] = freqz (b, a, N) where the frequency response is evaluated at N points equally spaced around the upper half of the unit circle (0 to $\pi\ rad$) and w is N frequency values.

[H, w] = freqz (b, a, N, ’whole’) - Evaluated over (0 to $2\pi\ rad$).




```
b = 1;
a = [1, -0.9];
N = 100; % number of samples
% calculation of frequency response
% w = [0:1:N-1]*pi/N - N frequency points

figure
subplot 211
[H, w] = freqz(b, a, N);
mag_H = abs(H); % magnitude spectrum 0 to pi
plot(w, mag_H);
xlim([0 2*pi]);
ylabel('Magnitude');
title('Magnitude spectrum - 0 to pi’);

subplot 212
[H, w] = freqz(b, a, N,"whole");
mag_H = abs(H); % magnitude spectrum 0 to 2*pi
plot(w, mag_H);
xlim([0 2*pi]);
ylabel('Magnitude');
xlabel('frquency in rad');
title('Magnitude spectrum - 0 to 2*pi');
```

Topic 2: Convolution and Filtering

Note

- ✓ In MATLAB 'filter' command is available to solve difference equation, given the input and the difference equation coefficients. In its simplest form this function is invoked by, **y = filter (b,a,x)** where b is the numerator coefficients array and a is the denominator coefficients array and x is the input sequence array. The 'filter' command can be used for FIR or IIR systems.
- ✓ In MATLAB , **conv** command takes two input sequences (e.g. input x[n] and impulse response h[n]) and returns their convolution as output. 'conv' command can only be used for FIR filters and not for IIR filters
- ✓ **conv (h,x)** and **filter(h,1,x)** would give the same result. The '1' represents that the denominator coefficient is one (i.e the system is FIR filter)
- ✓ Note that **filter** command returns a result that equals the input length. While the results of **conv** are longer (i.e. length(x) + length (h) -1). However, over the input length, the result of **conv** and **filter** are identical.

Example 8

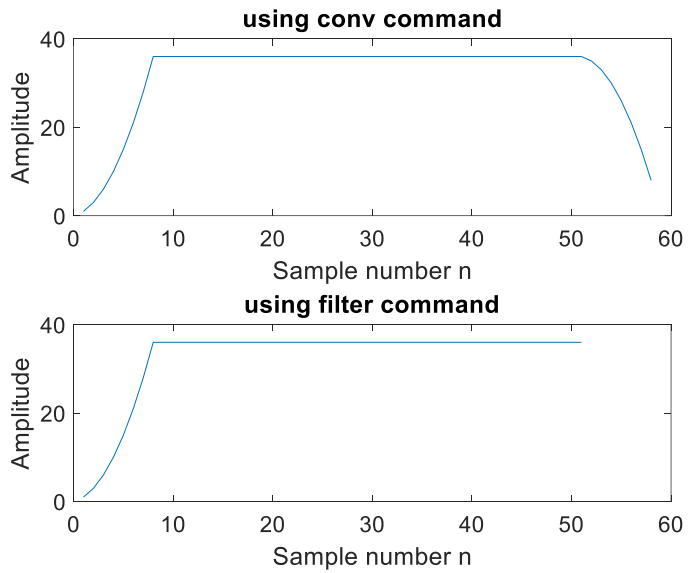
The difference equation describing a FIR digital filter is given by

$$y[n] = x[n] + 2x[n-1] + 3x[n-2] + \cdots + 8x[n-7]$$

Find the response $y[n]$ to the input $x[n] = \cos(2\pi n)$.

```
n = 0:50; % create a discrete index
x = cos(2*pi*n); % input signal
h = 1:8; % FIR filter coefficients from the above equation
a = [1 zeros(1,7)]; % denominator coefficient = 1
% Finding output response using filter command
yo = filter(h, a, x);
% Finding the response using convolution
yc = conv(h, x);
lyo = length(yo); % Length of yo
lyc = length(yc); % Length of yc
figure
subplot 211
plot(yc); xlabel('Sample number n');
ylabel('Amplitude');
title('using conv command');
subplot(212)
plot(yo); xlabel('Sample number n');
ylabel('Amplitude');
title('using filter command');
```

(see figure below– first 50 values are exactly same)



Topic 2: Discrete Fourier Transform (DFT)

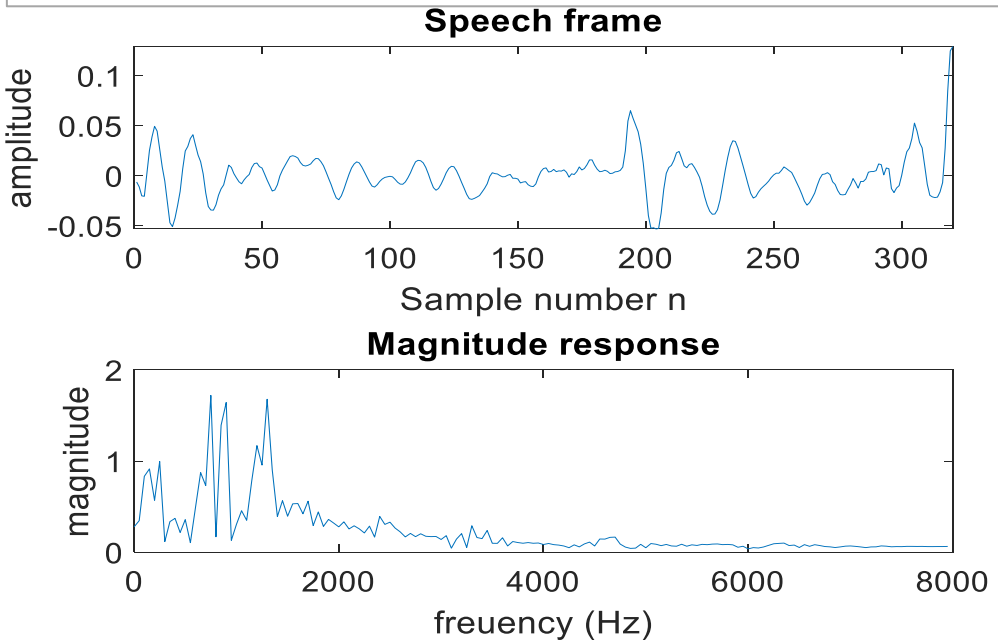
Example 9

A speech signal ('test_speech.wav') sampled at 16000 samples/sec is located in Moodle folder Resources/ Wav files for DSP Labs .

i. Read this file using 'audioread' MATLAB command.
ii. Write a **function** to compute the magnitude response of a speech frame of 320 samples.

For each frame :

a. Display the waveform
b. Display the magnitude response




```
% Read the audio file, fs = sampling frequency
[audio, fs] = audioread("test_speech.wav");
% Frame length in samples
frame_len = 320; % 20 ms frame
% Determine number of frames in the wav file
frame_num = floor(length(audio)/frame_len);
% Calculate and display for each frame
for i = 1:frame_num
        % framing the signal into 20 ms frames
        framed_signal = audio((i-1)*frame_len + 1:(i-1)*frame_len + frame_len);
        % calling the function
        mag_resp = magnitude_response(framed_signal);
        subplot 211
        plot(framed_signal); xlabel('sample number n'); ylabel('amplitude');
        xlim([0,frame_len]);
        title('Speech frame');
        subplot 212
        freq = (0:frame_len -1) * (fs/frame_len);
        plot(freq(1:frame_len/2),mag_resp(1:frame_len/2));
        xlabel('freuency'); ylabel('magnitude');
        title('magnitude response');
        pause;
end
```

```
function [frame_based_mag_response] = magnitude_response(framed_signal)
        % This function calculate the magnitude response of the given signal
        frame_based_mag_response = abs(fft(framed_signal));
end
```

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems
MATLAB Exercises

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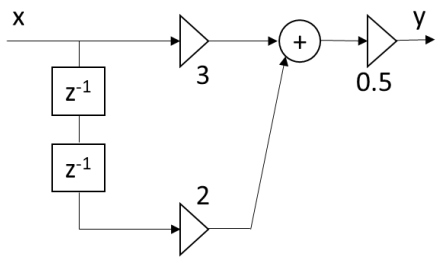
Term 3, 2026

Exercises – Discrete Time Systems (Topic 2)

Exercise 1 - Convolution

Write a MATLAB code to compute the output y of the system given below when the input $x = [1\ -3\ 2\ 4\ -2\ 2\ 6\ 2\ 4\ -2]$. Use the following MATLAB commands to compute y and compare the results and discuss it with your demonstrator.

a. 'conv' command
b. 'filter' command



Exercise 2: Pole-Zero plot

The transfer function of a discrete-time system is given by

$$H(z) = \frac{z^8 - (0.95)^8}{z^7(z - 0.95)}$$

Using MATLAB,

a. Find all the poles and zeros of the system and plot them in the z-plane
b. Compute and plot the magnitude response of the system
c. Compute and plot the impulse response of the system

Hint: The in-built functions *freqz* and *impz* could be used to compute the frequency and impulse responses

Exercise 3 - Impulse and Magnitude Responses

A 2nd order system has two zeros at the origin and two poles as $p_1 = re^{j\pi/4}$, $p_2 = re^{-j\pi/4}$, described by the following transfer function

$$H(z) = \frac{z^2}{(z - p_1)(z - p_2)}$$

a. Using MATLAB, compute and plot the magnitude response, impulse response, and pole-zero plane (z-plane) of three different 2nd order systems: (i) System 1: r = 0.8 (ii) System 2: r = 0.9 (iii) System 3: r = 0.99

b. Comment on the impact of the location of poles to the impulse response and magnitude response for three systems of (a)

Exercise 4 – Frequency Estimation using DFT/FFT

Given a discrete time sinusoid signal saved in a wav file '*signal1.wav*' located in Moodle folder Resources/ Wav files for DSP Labs. Using MATLAB,

a. Write a function *frequency_estimation* to read this wav file, and compute the DFT of the sinusoid signal and hence estimate its frequency.
b. How accurate your frequency estimation is? Discuss it with your lab demonstrator.

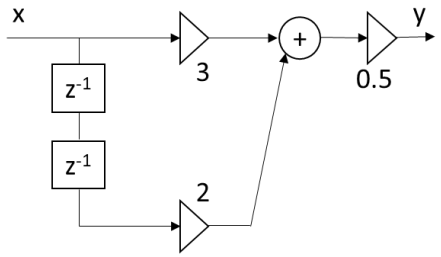
Hint: The in-built functions *audioread* could be used to read the wav file and signal sampling rate.

Exercises – Discrete Time Systems (Topic 2)

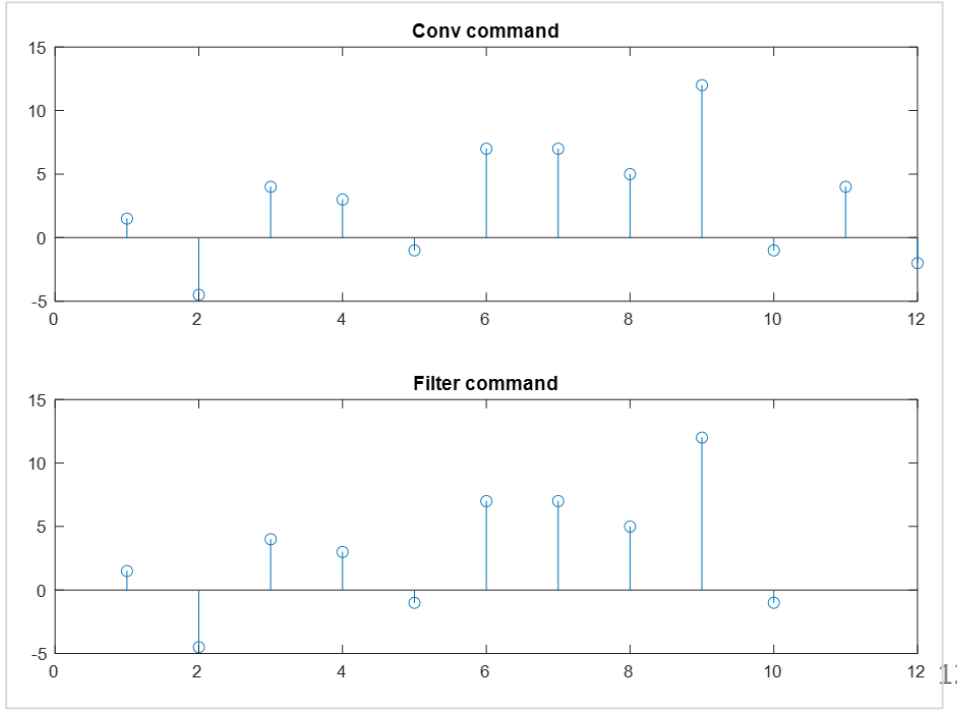
Exercise 1 - Convolution

Write a MATLAB code to compute the output y of the system given below when the input *x = [1 -3 2 4 -2 2 6 2 4 -2]* . Use the following MATLAB commands to compute y and compare the results and discuss it with your demonstrator.

a. 'conv' command
b. 'filter' command




```
x = [1 -3 2 4 -2 2 6 2 4 -2];
% convolution of x and the coefficients [3 0 2]
yC = conv(x, [3 0 2])*0.5;
% FIR filter: 1 is the denominator coefficient
yF = filter([3 0 2]*0.5, 1, x); % FIR filter
figure
subplot 211; stem(yC); title('Conv command'); xlim([0,length(yC)])
subplot 212; stem(yF); title('Filter command'); xlim([0,length(yC)])
% Display the lengths of the outputs for comparison
disp(['Length of yC: ', num2str(length(yC))]);
disp(['Length of yF: ', num2str(length(yF))]);
% As discussed in topic 2 examples the length of yC is not equal to the yF.
```



Exercises – Discrete Time Systems (Topic 2)

Exercise 2: Pole-Zero plot

The transfer function of a discrete-time system is given by

$$H(z) = \frac{z^8 - (0.95)^8}{z^7(z - 0.95)}$$

Using MATLAB,

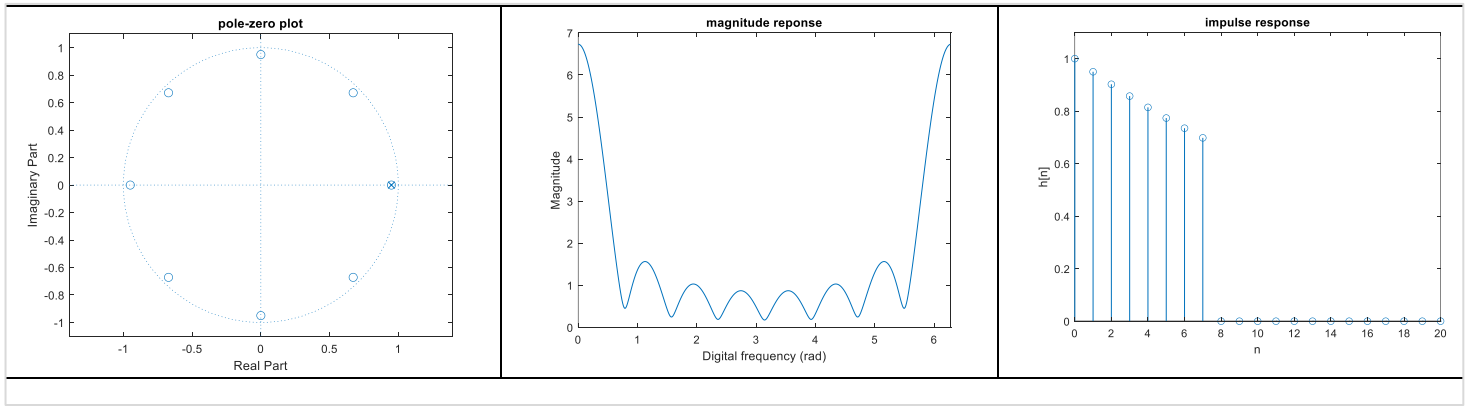
a. Find all the poles and zeros of the system and plot them in the z-plane
b. Compute and plot the magnitude response of the system
c. Compute and plot the impulse response of the system

Hint: The in-built functions *freqz* and *impz* could be used to compute the frequency and impulse responses

```
b = [1 zeros(1,7) -0.95^8]; % numerator coefficients
a = [1 -0.95 zeros(1,7)]; % denominator coefficients
zero = roots(b); % finding the roots of numerator
poles = roots(a); % finding the roots of denominator
figure
zplane(zero, poles); % pole-zero plot

figure
[H,w]=freqz(b,a,'whole'); % frequency reponse
plot(w, abs(H)); % plot magnitude reponse
xlim([0,2*pi]);
xlabel('Digital frequency (rad)');  ylabel('Magnitude');

figure
h = impz(b,a); % impulse response
stem(0:length(h)-1,h);
xlabel('n'); ylabel('h[n]'); axis([0 20 0 max(h)+0.1]);
```



Exercises – Discrete Time Systems (Topic 2)

Exercise 3 - Impulse and Magnitude Responses

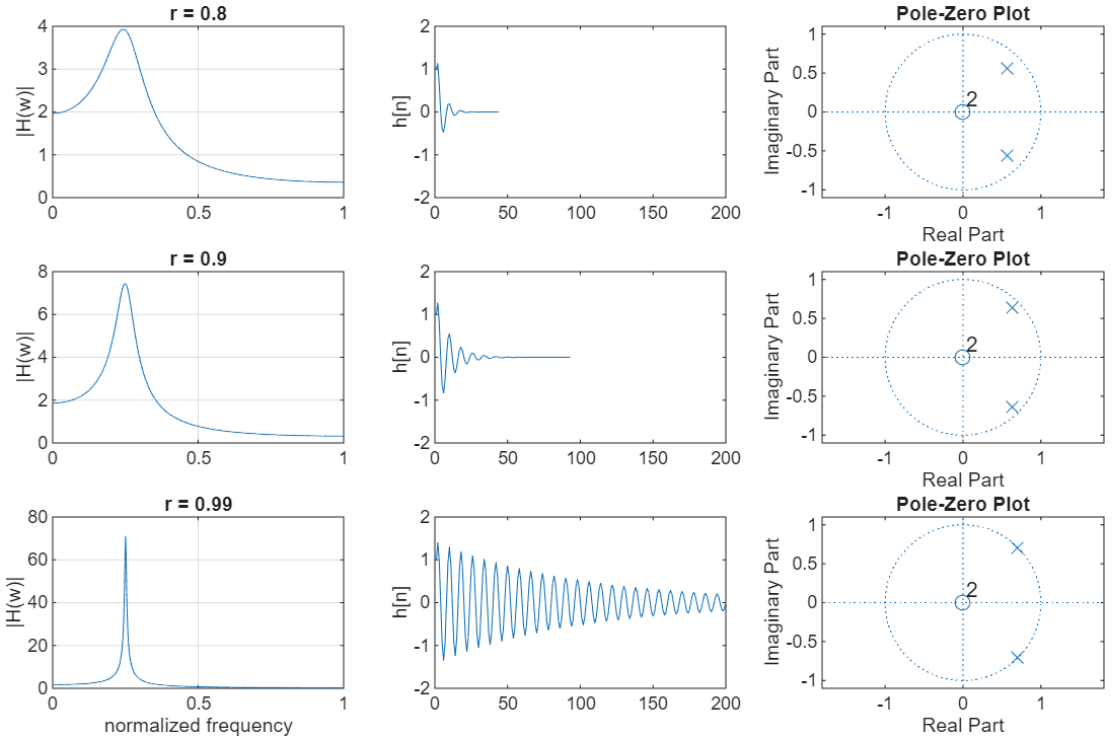
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$$H(z) = \frac{z^2}{(z-p_1)(z-p_2)}$$

a. Using MATLAB, compute and plot the magnitude response, impulse response, and pole-zero plane (z-plane) of three different 2nd order systems: (i) System 1: r = 0.8 (ii) System 2: r = 0.9 (iii) System 3: r = 0.99

b. Comment on the impact of the location of poles to the impulse response and magnitude response for three systems of (a)

```
theta = pi/4; % pole position
b = 1; % numerator coefficient 1
rs = [0.8, 0.9, 0.99]; % radius of three systems
figure
for i=1:length(rs)
        r = rs(i); % select radius from rs vector
        a = [1 -2*r*cos(theta) r^2]; % denominator coefficient
        [H, w] = freqz(b,a); % frequency response
        subplot(3,3,1+(i-1)*3); % 3*3 subplots
        plot(w/w(end), abs(H)); % x axis is normalised to 1
        ylabel('|H(w)|');
        title(['r = ' num2str(r)]);
        grid on
        subplot(3,3,2+(i-1)*3);  plot(impz(b,a)); % impulse response
        ylabel('h[n]');
        axis([0 200 -2 2]);
        subplot(3,3,3+(i-1)*3);  zplane(b,a); % pole zero plot
end
subplot 337 % select 7th plot and label the axis
xlabel('normalized frequency')
```



Exercises – Discrete Time Systems (Topic 2)

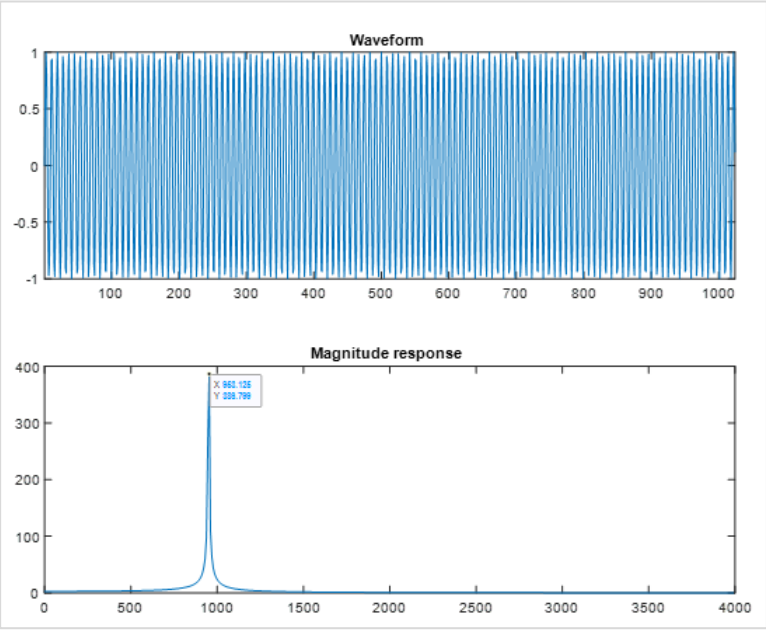
Exercise 4 – Frequency Estimation using DFT/FFT

Given a discrete time, sinusoid signal saved in a wav file *'signal1.wav'* located in Moodle folder Resources/ Wav files for DSP Labs. Using MATLAB,

a. Write a function *frequency_estimation* to read this wav file and compute the DFT of the sinusoid signal and hence estimate its frequency.

b. How accurate your frequency estimation is? Discuss it with your lab demonstrator.

Hint: The in-built functions *audioread* could be used to read the wav file and signal sampling rate.




```
function est_freq=frequency_estimation(wavFile)
        % read the wav file (sampling frquency fs = 8000)
        [x,fs]=audioread(wavFile);
        No_samp=1024;
        freqHz=(0:No_samp-1)*fs/No_samp; % freuency in Hz
        % frequency points upto half sampling frequency (fs/2)
        freqHz=freqHz(1:No_samp/2);
        xs=x(1:No_samp); % select 1024 samples from x array
        X=fft(xs,No_samp); % perform fft
        X_mag=abs(X(1:No_samp/2)); % obtain magnitude of FFT
        i=2; % set initial value to 2
        % search for a magnitude peak in a loop
        while i<length(X_mag)-1
                if (X_mag(i)>X_mag(i-1))&&(X_mag(i)>X_mag(i+1))
                peak_id=i; % set the peak_id index
                break % break the loop if conition met
                end
                i=i+1; % increment the loop counter
        end
        % use the peak id as an index to locate the
        % frequency and find the value
        est_freq=freqHz(peak_id);
        % plot routine
        figure
         subplot 211 ; plot(x(1:No_samp));  xlim([1 No_samp]); title('Waveform');
        subplot 212; plot(freqHz,X_mag); title(Magnitude response');
end
est_freq = frequency_estimation('signal1.wav');
```